COMP 693 Industry Project Weekly Journal

Project Details	
Student Name	Sneha Kuriakose
Student ID	1166658
Project Title	Air Quality Risk Predictor
Project Type	Independent Project

Week 3: 27 July 2026 – 2 August 2026			
Tasks Completed and Time Spent on each Task	Project setup (GitHub repository, virtual environment, project structure) Time : 2 hours Studied Open-Meteo and OpenAQ API documentation Time : 3 hours Developed Python script to collect historical weather data from Open-Meteo Time : 4 hours Developed Python script to collect historical air quality data from OpenAQ Time : 6 hours Debugging API authentication, timestamp formatting, and rate limit issues Time : 4 hours		
Project Hours this week	19 hours	**Sum of Total Project Hours (incl. prior week's hours)**	22 hours
Plans/Goals for Next Week	• Clean the weather and air quality datasets. • Handle missing values and duplicate records. • Merge the datasets using timestamps. • Convert pollutant measurements into air quality risk categories (Good, Moderate, Unhealthy.)		
Reflections	This week focused on building the data collection pipeline, which forms the foundation of the project. Although I expected data collection to be straightforward, I discovered that working with real-world APIs requires careful attention to authentication, timestamp formatting, and API usage limits. I learned how to read API documentation, process JSON responses, and convert them into structured datasets using Pandas. The biggest challenge was the OpenAQ API, where requests initially failed due to authentication requirements and rate limits. After investigating the documentation and testing different approaches, I implemented API key authentication, corrected the timestamp format, limited		

	the number of monitoring stations, and added delays between requests. Successfully solving these issues strengthened my debugging skills. By the end of the week, I had successfully generated weather and air quality datasets
Project Health	*Green* The project is progressing according to the planned timeline. The primary objectives for this week were completed successfully.
Issues Arising	• OpenAQ required an API key for authentication. • Incorrect timestamp formatting initially caused API requests to fail. • Requesting data from many monitoring stations resulted in API rate-limiting
Strategy to Resolve Project Health and Issues Arising	Not Required

Week 4: 03/08/2026 to 09/08/2026	
Tasks Completed and Time Spent on each Task	• Implemented the AQI labeling logic (src/labeling.py) using official EPA breakpoint tables for PM2.5, PM10, and NO2 Time :3 hours • Built and ran a demo script (demo_week2.py) to validate the labeling logic against four official EPA reference values Time: 2 hours • Reviewed fetch_air_quality.py in detail and identified that the documented 'auto' fallback mode (switching to Open-Meteo if no OpenAQ station is found) is not yet fully implemented Time :1 hour • Identified that the PM2.5 breakpoints currently used are the pre-2024 EPA table and need updating to the current standard Time: 1 hour • Updated the project README to document Week 2 progress Time: 1 hour • Researched other openly available data sources that could improve prediction accuracy such as traffic density and wildfire/fire activity data Time : 2 hours • Started learning Streamlit, for the planned web application interface Time: 3 hours

Project Hours this week	13 hours	**Sum of Total Project Hours (incl. prior week's hours)**	35 hours
Plans/Goals for Next Week	• Merge the weather and pollution datasets into a single combined dataset per city • Feature engineering — add other necessary columns to the merged table • Target output: one row per hour, combining weather and pollution information, covering the past year		
Reflections	What has gone well? Validating the AQI calculation against official EPA reference values gave confidence that the core labeling logic is mathematically correct. What has not gone as expected? Found that the PM2.5 breakpoints in use are outdated (pre-2024 EPA standard), and that a fallback feature described in the code's documentation isn't actually implemented yet		
Project Health	Green		
Issues Arising	Two known limitations identified: outdated EPA breakpoints for PM2.5, and an incomplete data-source fallback path. Neither is currently blocking progress, since the pipeline is running in a safe, working configuration.		
Strategy to Resolve Project Health and Issues Arising	Not required this week		

Week 5: 10 August 2026 – 16 August 2026	
Tasks Completed and Time Spent on each Task	• Built src/merge_clean.py — the core pipeline step that joins the weather and air-quality datasets on date time, removes duplicate timestamps, resamples both to hourly frequency, and interpolates small gaps Time: 4 hours • Added pollutant cleaning logic — removing negative/invalid sensor readings and clipping extreme outliers using the IQR method — before merging with the weather data Time: 3 hours • Built demo_week3.py to run the pipeline live and print a before/after comparison: raw, separate datasets in, cleaned/merged/labelled dataset out Time: 2 hours

	Time: 2 hours • Reviewed the merged output for both cities and found that Auckland's data is 100% labelled "Good" with no risk variation at all, while Delhi's spans the full risk scale Time: 2 hours		
Project Hours this week	15 hours	**Sum of Total Project Hours (incl. prior week's hours)**	50 hours
Plans/Goals for Next Week	• Run the pipeline for the remaining NZ cities (Wellington, Hamilton, Christchurch) • Start feature engineering • Begin training the model on the merged dataset		
Reflections	Successfully built and validated the merge-and-clean pipeline, and confirmed it runs correctly end-to-end for two very different cities.		
Project Health	Green		
Issues Arising	NZ air quality data currently shows not much variation in risk category (only "Good" observed for Auckland), which could limit the model's ability to learn to recognise higher-risk conditions if NZ data is used alone.		
Strategy to Resolve Project Health and Issues Arising	Not Required		